（令和７年度前期日程）

英　　語

90　分

注　意　事　項

1. 試験開始の合図までこの冊子を開かないこと。
2. 本問題冊子は16ページ，答案用紙は２ページである。
3. 各答案用紙の右上の枠内には，**受験番号**を記入し，左上段の枠内には，受験番号の**下２桁**の数字を忘れずに記入すること。
4. 解答はすべて各答案用紙の所定欄に明瞭に記入すること。
5. 問題冊子および答案用紙は切りはなさないこと。
6. 答案用紙に記入する受験番号の数字の字体は，下記の例にならい，明瞭に記入すること。

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試験問題は，つぎのページより始まります。

I 次の英文を読んで，以下の設問に答えよ。(80 点)

It seems quite certain that humans are the only creatures to develop a formal system of measurement. We know that many species, from rats to racoons, have an understanding of quantity and are able to distinguish between bigger and smaller piles of food, for example, while other animals carry out feats that surely require some intuitive form of reckoning (think about the astounding journeys made by birds across continents, orientating themselves using methods not fully understood by science). [A], and indeed, studies of children suggest that measurement, like writing and counting, is a cultural skill rather than an intuition we are born with.

In one study from 1960, children between the ages of four and ten were shown an 80-centimetre tower of blocks on a table and asked to build a tower of identical size a little distance away. While they were working on the task, a screen was placed between their tower and the original, stopping them directly comparing the two. To solve this puzzle, the youngest children, no older than four or five years old, eyeballed the job. They looked at the first tower, then started building their own approximate copy. The next cohort, children up to the age of seven, decided that visual comparison was not enough and used their bodies as measuring sticks. They held up their arms, hands, and fingers to the two towers, and compared their heights against their own. (At this age, some children also decided, quite understandably, that the game was rigged and simply ignored the instructions and built their tower next to the original.) The oldest group, children aged seven and over, turned to external measures, using strips of paper and sticks that had been provided as makeshift rulers. And even among these there were subtle differences. Younger children were happiest using paper and sticks the same height as the towers, while older individuals were comfortable using smaller items that could be counted as subdivisions of the whole.

Studies like this suggest not only that measurement is a skill we acquire with age, but that a key component of this practice is our ability to abstract. (1)単純に，ある塔と別の塔を比べたり，対象物と同じ高さの計測道具を使ったりするだけでは十分ではない。 We must instead create an intermediary: a unit of measure that represents nothing but its own value and provides a convenient medium for transferring information from one domain to another.

There's some suggestion that our ability to process number in this way is part of (2)a larger cognitive trade-off made long ago in our evolutionary past. Evidence comes from our closest genetic relative, the chimpanzee, who displays a remarkable facility with certain sorts of number tasks. With the right training, a chimp can be shown the numbers 1 to 10 on a screen for just a fraction of a second before tapping the now hidden digits in the correct order, doing so much faster and more accurately than humans. In fact, chimps can complete this task even if the numbers only appear for as little as 210 milliseconds. This is less time than it would take your eye to move around the screen, which suggests the skill being called upon is not a grasp of number as we understand it, but eidetic memory*: the ability to retain complex visual information after brief exposure. It's a fantastical ability, the sort we might associate with savants, but it has its own limitations. The same chimps that can perform these feats are unable to replicate other basic numeracy skills, like matching groups of items larger than four or five with the correct number, even after years of training.

Researchers behind this work theorise that some common ancestor of both chimps and humans possessed an eidetic memory, which would have been just the thing for identifying threats in a jungle environment. In a flash you could take in a tangle of leaves, vines, roots, bark, flowers, fruits, and *teeth*, identifying a potential predator and raising the alarm. At some point, though, evolutionary forces nudged a group of our ancestors into trading their souped-up memory for other cognitive aptitudes, including, we think, the ability to process language; to socialise and learn from one another. These were the cognitive tools that would

allow measurement to flourish, helping to construct the systems that now sustain so much of modern life.

It was the nineteenth-century British physicist William Thomson, better known as Lord Kelvin, who offered one of the most memorable summaries of the contribution of measurement to human knowledge. 'When you can measure what you are speaking about, and express it in numbers, you know something about it,' said Thomson, 'but when you cannot express it in numbers, your knowledge is of a meagre and unsatisfactory kind; it may be the beginning of knowledge, but you have scarcely, in your thoughts, advanced to the stage of *Science*.'

Thomson's words epitomise a sort of metrological triumphalism: a confidence in the power of number to square the untidy mysteries of the universe and tame the unknown through calculation. (3) It is a reasonable belief given the history of the sciences, where accurate measure has time and time again proved itself a prerequisite for experiment and a spur to discovery. Thomson's own groundbreaking work in thermodynamics and electromagnetism relied heavily on just such accurate observations, but this connection can be traced back much further to the discipline of ancient astronomy, a profession that blends what we would now categorise as mysticism and empiricism.

In Babylonia, a Mesopotamian kingdom that emerged around 1894 BC, the gods were believed to be regular, if oblique, communicators, bestowing everything from animal entrails to the 'colour of a dog that urinates on a man' with divine import. But the heavens were thought to provide a particularly authoritative format, with the ubiquity and clarity of the stars and planets constituting a celestial PA system that broadcast far and wide. [B]. Or they might herald a time of peace, the birth of a long wished-for child, or the start of a profitable trading arrangement. To decode these messages, Babylonian astronomers created detailed records of celestial movements in the heavens, using the resulting tables to sieve out irregularities and, with them, the favour of

the gods. This was the kernel of what we now call the scientific method — a demonstration that accurate observations of the world could be used to forecast its future.

The importance of measurement in this sort of cosmic comprehension did not develop smoothly over the centuries. Indeed, in the Middle Ages in Europe, reckoning with hand and eye was sometimes seen as producing a rather shabby sort of knowledge, inferior to that of abstract thought. This suspicion was due to the influence of ancient Greeks in the era's scholasticism, particularly Plato and Aristotle, who stressed that the material world was one of unceasing change and instability, and that reality was best understood by reference to immaterial qualities, be they Platonic forms or Aristotelian causes. It would take the revelations of the scientific revolution to fully displace these instincts, with observations of the night sky once again proving decisive.

Consider, for example, the unlikely patron saint of patient measurement that is the sixteenth-century Danish nobleman Tycho Brahe. By most accounts Brahe was an eccentric, possessed of a huge fortune (his uncle Jørgen Brahe was one of the wealthiest men in the country), a metal nose (he lost the original in a duel), and a pet elk (which allegedly died after drinking too much beer and falling down the stairs of one of his castles). After witnessing the appearance of a new star in the night sky in 1572, one of the handful of supernovae ever seen in our galaxy, Brahe devoted himself to astronomy. Ancient wisdom and religious doctrine held that the heavens were immutable, but Brahe castigated such 'blind watchers of the sky' — the new star plainly proved otherwise, and he spent decades compiling detailed and precise astronomical records in a purpose-built observatory named Uraniborg. It was the data collected here that would allow Brahe's assistant and colleague, the visionary German astronomer Johannes Kepler, to derive the first mathematical laws of astronomy, the three laws of planetary motion, which correctly described the elliptical orbits of the planets and treated the occupants of our solar system as ordinary matter rather

than divine or ethereal substance. It was measurement, then, that focused this new sort of attention on the cosmos, uncovering eternal truths that would eventually displace the verities of the Church.

By the time we reach Lord Kelvin, [C]. In the nineteenth century, it was precision engineering that transformed the steam engine from a leaky and inefficient machine to the high-pressure muscle of the Industrial Revolution, while the ability to accurately measure and meter electricity allowed for its commercial application in lighting, communication, and more. This was the century in which the factory replaced the farm as the backbone of a nation's wealth; in which telegraph lines connected continents and X-rays illuminated the interior of the human body. The 1800s may have begun under the flicker of torch and gaslight, but they ended with a blaze of electricity — all advances that owed at least something to measurement, and would only accelerate in the years that followed.

The calculations of Kepler are sometimes thought of as the first 'natural laws' of science, in that they are unchanging, precise, and verifiable. [D]: their predictions are applicable not just to *this* planet at *this* time but to all planets across time and space. In other words, they are generalised, abstract rules — qualities essential to the development of measurement. Indeed, if you were to summarise the history of measurement in a single sentence, it would be as a history of increasing abstraction. (4) Measurement begins life rooted in the particulars of human experience but over time has become increasingly detached from our life and labour. Just as with Kepler's laws, the result is that it has attained authority over an ever-expanding domain.

As the children who measured towers in that 1960 study demonstrated, the first tools of measurement we turn to are our own bodies. Common units like the hand and foot are still in use today, while others are at least familiar, like the cubit (the length from the elbow to the fingertip) and fathom (the span of your outstretched arms). Some cultures have created particularly rich indexes of

measure from the human body. The Aztecs not only had equivalents of the cubit and fathom, but also units based on the forearm alone (the *omitl*), the tip of the hand to the armpit (the *ciacatl*), and the tip of the fingers to the shoulder (the *ahcalli*). The Māori have derived at least twelve units from the body, from the smallest, the *konui*, equal to the first joint of the thumb, to the largest, the *takoto*, which measures the full length of the body with both arms raised above the head. [E]. We may have forgotten the *yepsen*, for example, a Middle English unit equivalent to a pair of cupped hands, but we still dole out food and ingredients by the pinch and mouthful.

[Adapted from James Vincent, *Beyond Measure: The Hidden History of Measurement.* Faber & Faber, 2022, pp. 7-13.]

* eidetic memory　直観像記憶

I -1. 下線部(1)を英語に訳せ。

I -2. 下線部(2)は具体的にどのような進化の過程を指しているか。チンパンジーと人間の能力の違いを踏まえつつ，85 字以内の日本語で答えよ。(句読点も文字数に含める。)

I -3. 下線部(3)を日本語に訳せ。

I -4. 下線部(4)を日本語に訳せ。

I -5. [A] から [E] のそれぞれに入れるのにもっとも適切な語句を選び，1から5の番号で答えよ。ただし，本文では大文字で始まるものも小文字で始まっており，同じ選択肢を2度使うことはない。

1. changes in the night sky might warn of impending disaster: of disease, flood, or invasion
2. but these skills are limited in scope
3. their authority is a product of their universality
4. many of these measures have been superseded by standardised units, but survive in an informal sense
5. the power of measurement has proven its supremacy not just through scientific knowledge but also its industrial application

I -6. 以下の①と②の答としてもっとも適切なものをそれぞれ1から5の中から選び，番号で答えよ。

① Which is the most likely reason that the author mentions William Thomson?

1. to argue that knowledge can be obtained through a variety of methods
2. to call out Thomson's bias for valuing science over other intellectual pursuits
3. to highlight that recording precise observations with numbers is critical in science
4. to marvel that it took until the 1900s for humans to appreciate numerical measurement
5. to persuade that Thomson should be remembered as the father of modern science

② Select the correct statement related to Tycho Brahe.

1. Brahe believed that God called him to dedicate his life and resources to the study of astronomy.
2. Brahe competed with Johannes Kepler to establish the first mathematical laws of astronomy.
3. Brahe continued to sponsor young scientists at Uraniborg, even after losing his ability to see.
4. Brahe disagreed with the old, inherited idea that objects in outer space are unchanging.
5. Brahe tragically died in an accident soon after constructing an observatory he named Uraniborg.

I -7. 以下の1から10の文の中から本文の内容に一致するものを3つ選び，番号で答えよ。

1. The author argues that because many animals are able to tell the difference between a few and a lot, humans must also be born with this capacity.
2. Researchers in 1960 selected children from similar backgrounds to participate in their study as a way to control for cultural influences on measurement styles.
3. In the experiment carried out in 1960, researchers discovered that while children of all ages used some kind of measuring tool, older children tended to prefer smaller items as aids in measurement.
4. The author praises children who defied the experimenters' instructions in 1960 because they demonstrated independent thinking characteristic of the scientific revolution.
5. Plato and Aristotle found a sense of security and predictability through observing phenomena in the world around them, as this allowed them to identify stable patterns within change.
6. Like ancient Babylonian astronomers, Tycho Brahe kept detailed records of his observations, though he did not look to the stars for fortune telling.
7. The author suggests that exact measurement was a factor that contributed to the shift from agricultural to industrial society in the 1800s.
8. Civilizations where adult body sizes were relatively uniform were more likely to have relied on body-based measurements than those with greater variability in body sizes.
9. Ancient Māori and Aztec peoples, like ancient Greeks and Babylonians, excelled in the study of astronomy, and applied their findings to society.
10. Some medieval Europeans looked down upon reckoning with body parts, but various peoples have developed such ways of measurement.

II 次の英文を読んで，以下の設問に答えよ。(70 点)

It sounds like a miracle, but no great technological leaps were required. In a commercial lab on the outskirts of Helsinki, I watched scientists turn water into food. Through a porthole in a metal tank, I could see a yellow froth churning. It's a primordial soup of bacteria, taken from the soil and multiplied in the laboratory, using hydrogen extracted from water as its energy source. When the froth was siphoned through a tangle of pipes and squirted on to heated rollers, it turned into a rich yellow flour.

[A] But the scientists, working for a company called Solar Foods, were allowed to give me some while filming our documentary *Apocalypse Cow*. I asked them to make me a pancake: I would be the first person on Earth, beyond the lab staff, to eat such a thing. They set up a frying pan in the lab, mixed the flour with oat milk, and I took my small step for man. It tasted . . . just like a pancake.

But pancakes are not the intended product. Such flours are likely soon to become the feedstock for almost everything. In their raw state, they can replace the fillers now used in thousands of food products. When the bacteria are modified they will create the specific proteins needed for lab-grown meat, milk and eggs. Other tweaks will produce lauric acid — goodbye palm oil — and long-chain omega-3 fatty acids — hello lab-grown fish. The carbohydrates that remain when proteins and fats have been extracted could replace everything from pasta flour to potato crisps. The first commercial factory built by Solar Foods should be running next year.

The hydrogen pathway used by Solar Foods is about 10 times as efficient as photosynthesis. But because only part of a plant can be eaten, while the bacterial flour is *mangetout*, you can multiply that efficiency several times. And because it will be brewed in giant vats the land efficiency, the company estimates, is roughly 20,000 times greater. Everyone on Earth could be

handsomely fed, and using a tiny fraction of its surface. If, as the company intends, the water used in the process (which is much less than required by farming) is electrolysed with solar power, the best places to build these plants will be deserts.

We are on the cusp of the biggest economic transformation, of any kind, for 200 years. While arguments rage about plant- versus meat-based diets, new technologies will soon make them irrelevant. Before long, most of our food will come neither from animals nor plants, but from unicellular life. After 12,000 years of feeding humankind, all farming except fruit and veg production is likely to be replaced by ferming: brewing microbes through precision fermentation. This means multiplying particular micro-organisms, to produce particular products, in factories. [B] I can see some drawbacks. But I believe it comes in the nick of time.

Several impending disasters are converging on our food supply, any of which could be catastrophic. Climate breakdown threatens to cause what scientists call "multiple breadbasket failures", through synchronous heatwaves and other impacts. The UN forecasts that by 2050 feeding the world will require a 20% expansion in agriculture's global water use. But water use is already maxed out in many places: aquifers are vanishing, rivers are failing to reach the sea. The glaciers that supply half the population of Asia are rapidly retreating. Inevitable global heating — due to greenhouse gases already released — is likely to reduce dry season rainfall in critical areas, turning fertile plains into dustbowls.

A global soil crisis threatens the very basis of our subsistence, as great tracts of arable land lose their fertility through erosion, compaction and contamination. Phosphate* supplies, crucial for agriculture, are dwindling fast. Insectageddon threatens catastrophic pollination failures. (1) It is hard to see how farming can feed us all even until 2050, let alone to the end of the century and beyond.

Food production is ripping the living world apart. Fishing and farming are, by a long way, the greatest cause of extinction and loss of the diversity and abundance of wildlife. Farming is a major cause of climate breakdown, the biggest cause of river pollution and a hefty source of air pollution. Across vast tracts of the world's surface, it has replaced complex wild ecosystems with simplified human food chains. [C] Eating is now a moral minefield, as almost everything we put in our mouths — from beef to avocados, cheese to chocolate, almonds to tortilla chips, salmon to peanut butter — has an insupportable environmental cost.

But just as hope appeared to be evaporating, the new technologies I call farmfree food create astonishing possibilities to save both people and planet. Farmfree food will allow us to hand back vast areas of land and sea to nature, permitting rewilding and carbon drawdown on a massive scale. It means an end to the exploitation of animals, an end to most deforestation, a massive reduction in the use of pesticides and fertiliser, the end of trawlers and longliners. It's our best hope of stopping what some have called the "sixth great extinction", but I prefer to call the great extermination. And, if it's done right, it means cheap and abundant food for everyone.

Research by the thinktank RethinkX suggests that proteins from precision fermentation will be around 10 times cheaper than animal protein by 2035. The result, it says, will be the near-complete collapse of the livestock industry. The new food economy will "replace an extravagantly inefficient system that requires enormous quantities of inputs and produces huge amounts of waste with one that is precise, targeted, and tractable". Using tiny areas of land, with a massively reduced requirement for water and nutrients, it "presents the greatest opportunity for environmental restoration in human history".

Not only will food be cheaper, it will also be healthier. Because farmfree foods will be built up from simple ingredients, rather than broken down from complex ones, allergens, hard fats and other unhealthy components can be

screened out. Meat will still be meat, though it will be grown in factories on collagen scaffolds, rather than in the bodies of animals. [D] But food is likely to be better, cheaper and much less damaging to the living planet.

It might seem odd for someone who has spent his life calling for political change to enthuse about a technological shift. But nowhere on Earth can I see sensible farm policies developing. Governments provide an astonishing £560bn a year in farm subsidies, and almost all of them are perverse and destructive, driving deforestation, pollution and the killing of wildlife. Research by the Food and Land Use Coalition found that only 1% of the money is used to protect the living world. It failed to find "any examples of governments using their fiscal instruments to directly support the expansion of supply of healthier and more nutritious food".

(2) Nor is the mainstream debate about farming taking us anywhere, except towards further catastrophe. There's a widespread belief that the problem is intensive farming, and the answer is extensification (producing less food per hectare). It's true that intensive farming is highly damaging, but (3) extensive farming is even worse. Many people are rightly concerned about urban sprawl. But agricultural sprawl — which covers a much wider area — is a far greater threat to the natural world. Every hectare of land used by farming is a hectare not used for wildlife and complex living systems.

A paper in *Nature* suggests that, per kilo of food produced, extensive farming causes greater greenhouse gas emissions, soil loss, water use and nitrogen and phosphate pollution than intensive farming. If everyone ate pasture-fed meat, we would need several new planets on which to produce it.

Farmfree production promises a far more stable and reliable food supply that can be grown anywhere, even in countries without farmland. It could be crucial to ending world hunger. But there is a hitch: a clash between consumer and producer interests. Many millions of people, working in farming and food processing, will eventually lose their jobs. (4) その新しい過程はたいへん効率的だか

ら，それらが生み出す雇用はそれらが壊してしまう雇用とは比べものにならないだろう。

RethinkX envisages an extremely rapid "death spiral" in the livestock industry. Only a few components, such as the milk proteins casein and whey, need to be produced through fermentation for profit margins across an entire sector to collapse. Dairy farming in the United States, it claims, will be "all but bankrupt by 2030". [E]

While I doubt the collapse will be quite that fast, in one respect the thinktank underestimates the scale of the transformation. It fails to mention the extraordinary shift taking place in feedstock production to produce alternatives to plant products, of the kind pioneered in Helsinki. This is likely to hit arable farming as hard as cultured milk and meat production will hit livestock farming. Solar Foods thinks its products could reach cost parity with the world's cheapest form of protein (soya from South America) within five years. Instead of pumping ever more subsidies into a dying industry, governments should be investing in helping farmers into other forms of employment, while providing relief funds for those who will suddenly lose their livelihoods.

[Adapted from George Monbiot, "Lab-Grown Food Will Soon Destroy Farming — and Save the Planet." *The Guardian*, January 8, 2020. www.theguardian.com/commentisfree/2020/jan/08/lab-grown-food-destroy-farming-save-planet.]

* phosphate　リン酸塩

Ⅱ-1. 下線部(1)を日本語に訳せ。

Ⅱ-2. 下線部(2)を日本語に訳せ。

Ⅱ-3. 下線部(3)で筆者は，集約農業(intensive farming)と比べて，粗放農業(extensive farming)がどのような点でさらに悪いと述べているのか。70字以内の日本語で具体的に説明せよ。(句読点も文字数に含める。)

Ⅱ-4. 下線部(4)を英語に訳せ。

Ⅱ-5. [A] から [E] のそれぞれに入れるのにもっとも適切な文を選び，1から5の番号で答えよ。ただし，同じ選択肢を2度使うことはない。

1. I know some people will be horrified by this prospect.
2. Industrial fishing is driving cascading ecological collapse in seas around the world.
3. It believes that the American beef industry's revenues will fall by 90% by 2035.
4. Starch will still be starch, fats will still be fats.
5. This flour is not yet licensed for sale.

Ⅱ-6. 以下の問への答としてもっとも適切なものを選び，番号で答えよ。

Which of the following is least likely to be produced by Solar Foods's new technologies?

1. fried fish
2. green salad
3. hamburger
4. meat pie
5. spaghetti

Ⅱ-7. 以下の１から８の文の中から本文の内容に一致するものを２つ選び，番号で答えよ。

1. While lab-made flour could be an ingredient in almost any product in the near future, it is intended to be commercialized for pancakes.
2. Sufficient water sources are expected to be available through 2050, allowing the world to be fed without much difficulty.
3. Farmfree food is likely to succeed because the taste and flavors have been proved to be more favored by human beings than traditional agricultural and fishery products.
4. The author anticipates that newly developed farmfree food will be more beneficial for human health than food obtained through traditional methods.
5. The author, who has long called for political change, is disappointed that current government farm policies, on the whole, tend to result in harm to nature.
6. Unable to identify examples of government interventions that promoted well-being, the Food and Land Use Coalition declared their own research a failure.
7. The author criticizes RethinkX for exaggerating the impact that farmfree production will have on farmers growing crops in fields.
8. The author thinks that governments should pour money into supporting industry threatened by new technologies and innovations.